

A CENTURY OF PROGRESS IN PHARMACOLOGY

Quantifying the evolution and its
driving factors

Not confidential

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Confidentiality note

The document is not confidential. It can therefore be viewed online by anyone.

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ABSTRACT

Abstract

This report presents a quantitative and historical analysis of Pharmacology, based on an extensive dataset of over 300,000 scientific works published in the top 50 journals over the past century, using OpenAlex data and the Science of Science framework. The study applied network analysis, BERTopic topic modeling, and machine learning to quantify the field's evolution and identify factors influencing the impact of academic work.

The findings reveal significant structural changes in Pharmacology, including increased collaboration, globalization, and a rising number of references per work, consistent with broader scientific trends. Pharmacology is characterized by a particularly high rate of knowledge obsolescence, indicating substantial shifts in research topics over time. Citation distribution shows a dual pattern (exponential head, power-law tail), supporting the cumulative advantage theory (associated with the “Matthew Effect”). In terms of impact, collaboration alone is not a strong predictor; however, international collaborations and institutional or national prestige are correlated with higher diffusion.

Topic modeling uncovered diverse citation dynamics across subfields. While machine learning models demonstrated some predictive power for long-term impact—highlighting author prominence, institutional prestige, topic patterns, reference count, and work type (e.g., review or meta-analysis) as key features—overall predictive performance remained modest.

Résumé

Ce rapport propose une analyse quantitative et historique de la pharmacologie, fondée sur un vaste corpus de plus de 300 000 travaux scientifiques publiés dans les 50 meilleures revues au cours d'un siècle, en utilisant les données d'OpenAlex et le cadre de la Science of Science. L'étude mobilise l'analyse de réseaux, la modélisation thématique par BERTopic et des approches de machine learning pour quantifier l'évolution du domaine et identifier les facteurs influençant l'impact des travaux académiques.

Les résultats mettent en évidence des transformations structurelles majeures en pharmacologie, notamment une collaboration accrue, une internationalisation croissante et une augmentation du nombre de références par article, en cohérence avec les tendances scientifiques générales. La pharmacologie présente toutefois un taux particulièrement élevé d'obsolescence des connaissances, traduisant des changements marqués dans les thématiques de recherche au fil du temps. La distribution des citations apparaît duale (tête exponentielle, queue en loi de puissance), ce qui soutient la théorie de l'avantage cumulatif (souvent associée à “l'effet Matthieu”). Concernant l'impact, la simple collaboration n'apparaît pas comme un facteur déterminant, tandis que les collaborations internationales et le prestige institutionnel ou national sont corrélés à une diffusion plus large.

La modélisation thématique a révélé des dynamiques de citations variées entre sous-domaines. Si les modèles de machine learning offrent une capacité prédictive limitée quant à l'impact à long terme, ils identifient néanmoins comme variables clés la notoriété des auteurs, le prestige institutionnel, les motifs thématiques, le nombre de références et le type de publication (revue ou méta-analyse).

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INTRODUCTION

This report summarizes the research project I began at the Human and Social Data Science Lab (HSDSLab) of the Budapest University of Technology and Economics. The goal was to provide a comprehensive overview of the field of Pharmacology, using a dataset of more than 300,000 scientific works spanning over a century, and to compare this field with general findings from the Science of Science. The project was conducted independently, although I received valuable support from the HSDSLab.

The study initially focused on a single Pharmacology journal, the British Journal of Pharmacology, as a tribute to the journal, which will celebrate its 70th anniversary in 2025. This work was presented to Péter Ferdinandy, the editor-in-chief of the journal and vice-rector of Semmelweis University, Budapest.

Subsequently, the analysis was extended to the entire field of Pharmacology to provide a broader perspective. This report primarily focuses on this second part due to its wider relevance and the overlap between the two analyses (the BJP analysis is also available in the project's GitHub repository).

Research problem: How has the field of Pharmacology evolved over time, and which factors most strongly influence the impact of academic works within this discipline?

By work, we refer to a scientific article or, hypothetically, books, theses, and conference proceedings. The scope of our data sources will be discussed later in this study.

The concept of impact can be ambiguous. Here, we associate it with the recognition received from the scientific community, measured through citation counts or derived metrics. This framework is admittedly imperfect and is extensively discussed in the literature review section.

A factor is defined as a feature that influences impact. Establishing causal links is—as is often the case in the Social Sciences—challenging, as experiments cannot be conducted¹. Accordingly, most of the studies considered focus on observed correlations, while also discussing potential causal mechanisms and sources of bias.

Overviews addressing these classic Science of Science questions have previously been conducted across all disciplines [2], or within specific fields such as Physics [3] or Network Science [4]. However, no comparable studies exist in Pharmacology, with the partial exception of [5], which is limited to a topic modeling analysis of drug safety, a subfield of Pharmacology.

Personal contributions:

- Conducting an extensive literature review, both for the overview and for further developments the lab may undertake.
- Extracting a dataset of more than 300,000 scientific works spanning over a century.
- Carrying out a comprehensive analysis of this dataset to study the evolution of pharmacological research and the factors influencing its impact, using both classical approaches and network science tools.
- Performing topic-level research analysis with advanced Natural Language Processing methods.
- Investigating citation distribution in order to model the diffusion of scientific works, through mechanistic models and machine learning models.

[GITHUB REPOSITORY OF THE PROJECT](#)

¹“Assessing causality is one of the most needed future developments in SciSci” [1].

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LITERATURE REVIEW

Science of Science (SciSci) can be regarded as highly interdisciplinary, as it is connected to numerous fields, particularly sociology and epistemology, statistics and data science, and the study of complex systems—especially through network science. The quantitative perspective of this field has been studied for a long time, beginning with a pioneering work in 1926 [6] and followed by further developments in the 1960s [7].

This section is not intended to provide an exhaustive review of SciSci; rather, it focuses on the general framework of SciSci (3.1), the description and evolution of scientific research (3.2), and the factors influencing the diffusion of scientific work (3.3).

SciSci encompasses many other topics that will be only briefly discussed here. These notably include scientists' choice of a research problem (a classic exploitation–exploration dilemma), scientists' career dynamics (particularly mobility), credit allocation (how the credit given to a work by a community is distributed among its authors), the evaluation of research funding and the effects of peer review, and the study of interdisciplinarity (often contrasted with “in-silo” research). For a broader overview, we recommend [1].

GENERAL FRAMEWORK OF SCIsci

3.1.1 • *KNOWLEDGE UNITS*

The first challenge in studying knowledge diffusion is often to define the “knowledge unit” under investigation [8]. This unit can be an entire academic publication, a term (a single word or a group of words)² or a macro-level aggregation, such as an author, a discipline, or a journal.

Most studies equate concepts, ideas, and topics with terms, thereby adopting, to some extent, a Hegelian perspective³. The development of Natural Language Processing has greatly facilitated this approach.

3.1.2 • *ACADEMIC NETWORKS USED TO STUDY THE DIFFUSION*

Scientific research is a complex system that can be modeled using a multi-scaled hypergraph, first conceptualized by Latour and subsequently referred to as the Latourian network [9].

Mathematically, the Latourian network is simply a hypergraph—i.e., a graph in which connections can involve more than two nodes. It is important to note that this graph is heterogeneous: both nodes and edges can be of different kinds. For example, nodes can represent ideas, works, authors, countries, journals, or years. For the purposes of understanding and visualization, it is clearer to separate the different kinds of nodes into homogeneous layers; this is what we call a multi-layer graph [1], [9].

The three main layers studied are:

- Co-authorship layer: nodes are authors, and edges represent collaborations. This network is particularly important for studying collaboration patterns among researchers.
- Citation layer: nodes are works, and edges represent citations between works. This network is especially relevant for assessing the importance of a work.

²Some studies prefer to refer to “scientific memes”, by analogy with social media memes. A meme is then defined as a structure that, once introduced, is replicated in subsequent studies.

³Hegel's corresponding thesis is often summarized by the quote “it is in names that we think”.

- Idea layer: nodes are ideas, and edges quantify the co-occurrences of keywords within works. This network describes the evolution and proximity of topics.

These three layers interact with one another. However, due to the complexity of the full network, they are most often studied separately and independently. Such studies are therefore partial and imperfect. In contrast, [9] attempts to study the Latourian graph as a whole, in order to gain realism.

These layers are not the only useful representation. Other network can also be valuable, such as co-citation networks, in which two works are linked if they are cited together in a third study. This network is useful for measuring the proximity between works [1].

3.1.3 • *IMPACT METRICS*

We define impact as “diffusion within the scientific community”, or, more precisely, as “credit given by the academic community”. We do not equate impact with “success”, “merit”, “degree of innovation”, “importance for a theory or future research”, or “quality”. Within this framework, the impact of a work will be measured by the number of citations it receives, or by derived metrics. Citations do not perfectly capture credit, and alternative metrics (also known as altmetrics)—such as the number of views, downloads, or mentions on social media—are increasingly used and studied [1]. In this work, impact and diffusion will be used synonymously.

The number of citations reflects at least two distinct factors:

- the credit given by the scientific community,
- the practice of citing in research.

Identifying the best metrics for measuring the impact of a work, an author, or a journal has become a major challenge in the Science of Science; this subfield is often referred to as Scientometrics. These metrics have been extended to evaluate authors (e.g., the h-index) and journals (most notably the Journal Impact Factor, defined as the mean number of citations received by works published in the journal during the previous two years) [1].

It is important to note that—because of the second factor—substantial variation in citation counts can occur between fields, as well as across different time periods. The former is not an issue for this work, which focuses on a single field, but the latter may be problematic.

Older papers have had more time to accumulate citations than newer ones. This creates a significant age bias when comparing works over long periods. Several metrics have been proposed to address this issue. If we assume that the attention given to a paper does not decay over time (i.e., that the paper does not “age”), the citation rate $\frac{N_{\text{citations}}}{\text{Age}}$ can be informative. However, an aging effect is generally assumed, due to dilution or obsolescence phenomena. In such cases, the citation rate tends to penalize older papers excessively. A more accurate approach is the Mean Normalized Citation Score (MNCS), defined as $\frac{N_{\text{citations}}}{\overline{N}_{\text{citations, year, topic}}}$ where $\overline{N}_{\text{citations, year, topic}}$ is the mean number of citations for works published in the same year and field [10]. This score compares a work to its contemporaries in the same discipline and year of publication.

DESCRIPTION OF SCIENCE

The first two subsections will present the two main networks: the citation network (primarily used to study impact) and the co-authorship network (primarily used to study collaboration). The following two subsections will examine clusters derived from these networks, such as countries, institutions, or topics. The final subsection focuses on the temporal evolution of science.

3.2.1 • *DISTRIBUTION OF CITATIONS: THE CITATION NETWORK*

The global number of references made balances with the global number of citations received; however, their distributions are very different. [7] was the first to study the distribution of references and citations in academic

works. We will here focus on the citations distribution, but the references distribution have also been extensively studied by the literature.

3.2.1.1 - Citations distribution

[7] found that the distribution of citations follows a $\frac{1}{n^{2.5}}$ or $\frac{1}{n^3}$ law. Approximately 35% of papers are never cited, and 49% are cited only once.

[11] confirmed a $\frac{1}{n^3}$ law for the asymptotic tail of the distribution using a Zipf plot. For low-citation works, semi-log and log-log density plots reveal that neither an exponential nor a power law fits the data well. A stretched exponential distribution ($\exp\left[-\left(\frac{x}{x_0}\right)^\beta\right]$) fits reasonably well for $x \leq 200$ with $\beta \approx 0.4$. This study also found that 47% of papers are never cited.

Such results suggest that the diffusion of low- and high-cited papers may follow different mechanisms. For example, [11] conjectures that low-cited papers are primarily referenced in the future works of the same authors (self-citation) or their close collaborators, and are then quickly forgotten. By contrast, highly cited works diffuse through a broader collective interest and persist over a long period. [12] confirms that highly cited papers have a longer lifespan than low-cited ones. Subsections 4.2.1.3 and 4.2.1.4 explore possible mono- and dual-mechanism explanations.

As with many fat-tailed distributions, there remains uncertainty over whether the tail is best described by a power law or a log-normal distribution. [12] notably finds that the distribution fits better with a log-normal law, contrary to [7] and [11]. However, the data used in [12] have significant limitations (only citations within a single journal are considered), and the power-law hypothesis remains dominant.

Interestingly, the distributions of the number of collaborators per author (along all its career) and the number of papers per author follow a similar form (exponential $\times$ power law) [13].⁴

3.2.1.2 - Explaining the tail of the citations distribution

The distribution of the number of citations can be interpreted as the degree distribution of the citation network. This interpretation may help explain the mechanism underlying a power-law distribution. Indeed, the Barabási–Albert model of networks is well known to generate such distributions [15]. These networks are often referred to as scale-free in network science because:

- the degree distribution is dispersed (heavy-tailed), meaning that even very high degrees cannot be ignored, rather than being concentrated around a characteristic value as in most classical networks. There is no typical degree value;
- $\mathbb{P}(\alpha k) = \alpha^{-\gamma} \mathbb{P}(k)$ — when rescaled, the distribution remains exactly the same up to a multiplicative constant.

In the Barabási–Albert model [15], the network grows through the progressive addition of new vertices, which attach preferentially to already well-connected nodes. More precisely, the fraction of new citations going to existing works with k citations (the attachment rate) must be linearly proportional to k in order to yield a power-law distribution.

This constitutes the preferential attachment mechanism (in the network view), more generally referred to as cumulative advantage. The resulting effect—namely, the emergence of a power law—is called the Matthew Effect⁵.

Interestingly, [16]—23 years before the introduction of the scale-free network by Barabási and Albert—developed a different mechanism to explain the power-law citation distribution, using an urn model inspired by the Pólya urn process rather than a network paradigm. However, this formalization is more descriptive than explanatory, less precise regarding the dynamics of the citation network, and provides fewer insights into the underlying mechanism.

⁴Historically, the power-law nature of the distribution of Number of authors(Number of publications) is considered the first result in the Science of Science, discovered in 1926 by Lotka [14].

⁵Named in reference to a line from the Gospel of Matthew: “For to everyone who has, more will be given, and he will have abundance; but from him who does not have, even what he has will be taken away”. Also known as the “richer-get-richer” effect. This phenomenon is common; the Pólya urn experiment is a well-known example of a cumulative advantage process.

A major limitation of the preferential attachment framework is that it does not account for knowledge obsolescence. Papers that are successful for a time can become outdated as the field advances (knowledge obsolescence) [14], [17]. [17] also found that recent papers are forgotten more quickly than in the past, and that this attention decay is linked to the increasing volume of published works (knowledge dilution).

3.2.1.3 - Going through a model explaining the dual mechanism

Whereas the previous subsection focused on the distribution of the tail, some mechanisms aim to explain the entire citation distribution. [18] proposes the GGNR mechanism, described in the box below. This mechanism supports the consensus for a dual-mechanism explanation of the citation distribution. It formalizes two ways in which a researcher may discover a paper that they will subsequently cite:

- a direct way, in which the researcher finds the paper immediately (i.e., without intermediaries or proxies);
- an indirect way, in which the researcher discovers the paper by reading other “proxy” papers that reference it.

The indirect pathway becomes prominent once a paper surpasses approximately 25 citations. The results of this model fit both the exponential head of the distribution and its power-law tail.

Generalized Growing Network with Redirection (GGNR)

Hypothesis:

- All papers have the same number of references, n .
- At each time step, a new paper is added to the network; at a given time, the network has N nodes.

Let us consider A an existing paper, and denote k its number of incoming citations.

- The probability that the next paper added directly finds A is $r_{\text{direct}} = \frac{1}{N}$. This represents the direct mechanism.
- The new paper can also discover A through another existing paper B . There are nN references in the entire citation network, and k of them point to A . The probability that the new paper indirectly finds A is thus $r_{\text{indirect}} = \frac{k}{nN}$.

Once the new paper (more precisely, its authors) finds A , it may cite it directly or explore one of its references. Let c be the probability that the new paper explores one of A 's references instead of citing A directly.

If we denote $R(k)$ as the expected number of citations given from a new paper to A (i.e., the probability that a new paper cites A), we have:

$$R(k) = n \left[(1 - c)r_{\text{direct}} + cr_{\text{indirect}(k)} \right] = \frac{n(1 - c)}{N} + \frac{kc}{N}$$

Remark: As mentioned previously, Barabási and Albert [15] emphasized that defining the preferential attachment rate as a linear function of k is necessary to obtain a power law. This constraint is respected here and can thus be explained by this mechanism.

The proportion of works with k citations, $p(k)$, can then be deduced: $p(k) = p(k - 1)R(k - 1) - p(k)R(k)$ (flow entering - flow leaving the group of works with k citations).

From this, a general equation can be derived. $p(k)$ becomes equivalent to a power law for large k (using Stirling's approximation) and an exponential law for small k .

The uniform distribution of direct citations considered in [18] explains well the structure of citation distributions but is unrealistic, as it “means that in the initial period papers accumulate citation basically randomly” [19]. [19] proposes a direct citation distribution proportional to the team size of the work. The direct citation represents the initial attractiveness of the article; thus, the first diffusion of a work mainly depends on the authors' community (self-citation or low-distance citations). Using the general distribution of the number of authors per paper, the model is validated on real data and successfully reproduces the citation distribution.

It remains a structural model: its goal is to predict the global citation distribution more than predicting the success of individual works.

[19] adds that the indirect mechanism is probably linked to the sum of the “influences” gained by the papers citing the work, instead of just their number.

The age bias is not taken into account by [19]. This is not a problem because the dataset used in this study is composed of works published in a small period. This allows us to separate the obsolescence study from the citation mechanism study. For predictive purposes, however, training data would likely have very different publication years. One option would be to consider only the MNCS as a target⁶, predicting how the work performed compared to its competitors of the same publication year. Another option would be to integrate the age in the model.

Another limitation is the absence of the intrinsic value of works. In fact, the above model describe the global distribution rather than the impact trajectories of individual papers.

The age bias (due to the exponential attention decay given to a paper) can be tackled with the introduction of an aging factor, and the quality of a paper can be modeled with a fitness parameter [20]. [20] proposes a model using a different framework to include these two factors. Only the indirect mechanism seems to be considered. Using the same notations as for the GGNR, we have:

$$R(A(t)) \propto \eta_A k(A(t)) \left(\frac{1}{\sqrt{2\pi\sigma_i^2 t}} \exp \left[- \left(\frac{(\ln(t) - \mu_i)^2}{2\sigma_i^2} \right) \right] \right)$$

with η_A a fitness parameter and (μ_i, σ_i) describing the attention decay.

We recall the GGNR relation if we exclude the direct mechanism:

$$R(k) \propto \frac{k}{N}$$

The differences are:

- The introduction of a fitness parameter η_A .
- The introduction of the age of A, t , and the modeling of the exponential attention decay.
- The size of the network N is not used; however, if we suppose that this size grows linearly with the age of the work, the $\frac{1}{t}$ factor is equivalent to $\frac{1}{N}$. Remark: over long periods, the growth of the literature has been found to be exponential, but over the first year of the life of an article, the linear hypothesis may be credible.
- Only the GGNR indirect mechanism is considered in this new modeling.
- Finally, this new modeling allows prediction of individual papers, whereas the GGNR is only a structural model for predicting the citation distributions of all works.

We have been focusing here on the diffusion of works; however, the diffusion of ideas can also be studied to gain granularity. We highly recommend [8] for this kind of modeling.

3.2.2 • *AUTHORS AND COLLABORATION: THE COAUTHORSHIP NETWORK*

3.2.2.1 - Topology of the network

[13] found the coauthorship network of PubMed works (a biomedical database, therefore particularly relevant for our work) to be a small-world network. A small-world network is characterized by a small average path length and a strong clustering coefficient⁷. More precisely, its average path length is proportional to the logarithm of the size of the network. The coauthorship network is described as highly connected, with no risk of fragmentation.

Dynamically, when a field grows, the density of the coauthorship network increases, and the average path length (network diameter) of the main connected component decreases. A topological transition leads the network,

⁶Personal consideration. To the best of my knowledge, this approach has never been considered.

⁷“friends of my friends are often friends”

originally separated into many small components, to form a giant connected component. A parallel is made between this transition and the percolation phenomenon in Physics [13].

Metric	[13] / PubMed	[13] / All databases
Average path length	4.6	6
Diameter	24	-
Clustering coefficient	6%	30%
Authors per paper	3.7	-
Mean degree	18	-
Size of the giant component (% of number of authors)	92%	80%

Table 1: Topology of the coauthorship network

Better average path length but worse clustering coefficient is observed for medical sciences compared to the hard sciences. A low clustering coefficient is often due to a more hierarchical organization (i.e., a network with more tree motifs and fewer interconnections).

The structure of the coauthorship network naturally implies some clustering: all authors of studies with more than 2 authors are linked to each other. However, [13] finds that this explanation of the clustering is not sufficient, and that scientists probably tend to introduce pairs of their collaborators to other researchers.

[14] conducts a dual study of both the coauthorship network and the citation network. It confirms that authors of a publication tend to cite themselves preferentially (though this behavior does not increase over the years), and cite preferentially their collaborators instead of other authors. Self-citations are found to account for 20% of all citations over a long period. Self-citations occur earlier than citations from previous collaborators, which themselves occur earlier than citations by unrelated researchers (with a mean delay of 4.1 years for self-citations against 9 years for long-distance citations).

When A cites B, does B tend to cite A later, preferentially, perhaps as a form of “reciprocal acknowledgment”⁸? [14] finds a rate of reciprocal citations far higher than expected if this phenomenon did not exist. However, it does not fully answer the question, as it recognizes that this rate may simply reflect that if A cites B, they probably work on the same research topic. Therefore, the comparison must be made within the subnetwork corresponding to the topic of research; a comparison with the whole scientific network is irrelevant.

Interestingly, combining the citation network and the coauthorship network allows the study of the preference for self-citations, but not the preference for rewarding researchers who previously cited us. Addressing this question would require adding a third network, i.e., a topic network.

3.2.3 • GEOGRAPHIC AND INSTITUTIONAL DESCRIPTION

[21] found that citation and collaboration flows between countries are increasing. However, the strength of these flows still tends to decrease with distance, implying that the internet has not erased geography as a constraint in research. More precisely, the strength of the flows according to distance follows a power law, meaning that long-distance collaborations, while less frequent, are not rare (fat-tailed distribution). A gravity model can be applied and fits the observations. The total research impact of a country is found to grow linearly with national research funding. This study, however, does not assess differences in the diffusion of works between countries.

3.2.4 • TOPIC MODELLING

Assigning one or a few topics or representative keywords to each work in a database allows a major improvement in terms of granularity for studying knowledge diffusion. Some databases assign general topics to works,

⁸If we consider citations as a currency (analogy often made), a partial analogy is the notion of a cartel in micro-economics: producers/scientists making more or less explicit deals to increase their profits/citations

but these are often too broad to be informative, and many do not catalog the original keywords provided by the authors.

Classic topic modeling methods include Latent Dirichlet Allocation (LDA) and Non-negative Matrix Factorization (NMF). However, [22] developed BERTopic, based on embeddings and a BERT encoder, which achieves significantly better performance.

Once having topics assigned to the works, it is possible to study the spread of topics in the scientific community. This is often made using epidemiological modelling [23] or even economics analogy [24].

3.2.5 • EVOLUTION OF SCIENCE

The evolution of research is studied through several proxies. It is therefore important to recall that the observed evolution is a combination of two dynamics: the evolution of content and the evolution of citation practices [7].

Number of works. An exponential growth of the volume of works is observed [25], [14], with an average doubling time of 15 years. [14] also found an exponential rise in the number of distinct authors.

Number of references. [26], [14] report a steady increase in the number of references per work over 50 years. This rise can be attributed both to changes in citation practices and to longer articles.

Collaboration. A steady increase in the number of authors per paper is observed, from around 1 in the 1900s to about 4 in the 2000s [14]. Team size increased by 17% per decade over a 45-year period. Whereas in 1955 works in science and engineering were roughly 50% solo-authored and 50% team-authored, the ratio is now 10%-90% [27].

Number of citations. [14] observes a citation evolution characterized by rapid growth followed by significant decay for a set of Physics works. This evolution appears incompatible with the preferential attachment (PA) mechanism, which is likely only pertinent within subfields, but not for an entire corpus of scientific works (since new subfields emerge and others decline, a factor not accounted for by PA). The rate of uncited papers is also decreasing rapidly, from 80% in 1900 to 10% in medicine [28].

Ideas. The growth in the number of works cannot be equated with the growth of ideas. A linear expansion of new ideas (the “cognitive” expansion of science) is observed [25], measured through the lexical diversity of titles. Interestingly, small teams and solo authors appear to contribute more to cognitive expansion than larger teams, even though they often receive fewer citations.

Measuring the cognitive extension

[25] starts by cleaning the texts, replacing all uninteresting generic terms (“is”, “effect”, punctuation, etc.) with delimiters. These delimiters separate “phrases” (more often called tokens in NLP) that are assumed to represent scientific concepts. If the length of a phrase exceeds 3 words, only the last 3 words are kept, which tends to remove uninformative qualifiers and retain the core concept. The number of phrases is then normalized according to the length of the title.

Most other idea-level studies use similar methods [8].

FACTORS INFLUENCING THE DIFFUSION OF ACADEMIC WORKS

3.3.1 • ASSESSING THE CORRELATIONS BETWEEN POSSIBLE FACTORS AND THE DIFFUSION

Assessing the correlation between potential factors and our target is necessary for good modeling.

Other targets have also been studied. Notably, [26] studied the influence of different factors on the number of references in a work (e.g., does the academic age of a researcher or their productivity influence the age and number of references they use?).

Year of publication. The influence of the year of publication on impact is null when the MNCS is used as the target (giving the same weights to every year) instead of the raw number of citations.

Size and extension rate of the field. [29] finds that the larger a field of research, the slower its progress, theorizing an ossification of the knowledge network as the field grows. This phenomenon is linked with the Matthew effect and the formation of a “canon” of works that newcomers cite to enter the field. The effect is strengthened if the growth rate is high. Two mechanisms are proposed:

- Reviewers overwhelmed by the quantity of submissions are “forced to resort to heuristics” rather than assessing the quality of each paper in depth.
- Competition among too many new ideas prevents any of them from becoming widely known.

Number, age, and quality of references. In his 1965 classic work, [7], Da Solla Price already noted: “we know little about any relationship between the number of times a paper is cited and the number of bibliographic references it contains. [...] I conjecture that the correlation, if one exists, is very small”. [30] precisely described the relationships between the number of citations and the number of references, the number of citations of the references, or the age of the references. These relationships are non-linear, with inverted-U or inverted-L shapes, and plateaus observed, for example, between 20 and 50 references or between 5 and 10 years for the mean age of references. This suggests that these features primarily discriminate works with extreme characteristics, but do not differentiate the majority of works.

Co-occurrences of references. [31] studied reference patterns using a network linking two references if they appear in the same paper and compared the observed weight distribution with a randomized network to rank pairs of references from the most typical to the most atypical.⁹ They conclude that the most successful works are those that balance mostly typical combinations of references with a few atypical combinations. Works with too many atypical combinations or those too conventional perform worse.

Authors and institutions. The prestige of authors’ institutions significantly impacts the diffusion of a paper [32]. This difference persists at the country level; e.g., AI works with a US first author have significantly higher impact than those with a Chinese first author [32]. [33] also highlights a “diversity-innovation” paradox: women and minority-originated researchers are proportionally responsible for more new concepts but experience lower adoption of their works over time.

Collaboration. Team science is often found to have higher impact than solo-authored works. [27] finds that team works receive on average 2.87 times more citations than solo-authored works in medicine, after accounting for self-citation bias. This bias arises because each author can self-cite the work, increasing citations for team works. However, caution is needed: [27] uses raw citation counts rather than MNCS, and two phenomena must be considered:

- The mean number of citations varies strongly with the year of publication.
- Collaboration practices also vary strongly over time.

If team works are on average younger than solo-authored works, the difference may be biased by the citation variation across years.

[34] is particularly interesting because it studies the correlation between collaboration and quality using a dataset of works with expert review marks. No correlation is observed, suggesting that correlations with citations may be due to citation-specific biases.

A 2021 meta-analysis found a weak positive correlation of 0.146 but recognized a lack of consensus on the source of this correlation (quality or citation biases such as self-citation) [35].

⁹More precisely, they simplify and aggregate this massive network by looking only at the journals of the references.

Collaborations between authors from different research collectives [8] and international collaborations [32] are also often described as having higher impact on citation counts.

3.3.2 • CLASSIFYING THE POSSIBLE FACTORS

The goal is to list the factors that may influence our target (the diffusion). However, it is important to note that most of these factors are not directly measurable. Therefore, it will be necessary to find measurable features to evaluate them. Some measured features may be imperfect or influence not only one but several factors. Constructing a complete Direct Acyclic Graph of our theory to map all causal links is currently difficult due to the number of possible factors and metrics. However, based on the previous subsection, we propose a classification of the potential factors and their metrics and discuss this classification.

This subsection is therefore a theoretical development based on the literature described above. It is also largely derived from the classification proposed in [8] (at least for the internal part). We additionally drew inspiration from the Aristotelian classification (ethos/logos—pathos does not seem relevant here) of factors influencing the impact of a speech on its audience.

- **Structural** *Not linked to the work itself but to systemic mechanisms*

- The way a reader finds a paper and decides to reference it is a structural mechanism. It was developed in a previous subsection and successfully explains the global distribution of citations.
- This mechanism requires hypotheses on the number of references per work, size of the field, and speed of the field volume extension.
- The rate at which a work becomes outdated, the obsolescence factor, and the rate of self-citations must also be accounted for.
- These factors naturally depend on the year and the field type, which are therefore also structural factors.

- **Individual** *Specific to the work.*

→ **ethos** (legitimacy/confidence/importance that will be given to the work just because of its metadata, without really reading it)

- Background of authors (prestige, prominence, origin...)
 - Mean number of past works of authors (= academic age × mean productivity)
 - Mean MNCS of authors at the time of publication
 - Gender and social origin of the authors (if any form of discrimination exists)
- Institutions of the authors
 - Aggregation of the metrics computed for authors
- Degree of collaboration
 - Number of authors
 - Number of distinct institutions/countries

→ **logos** (linked to the content of the work itself)

- Pragmatic usefulness (= innovation × usefulness of this innovation for future research)
- Comprehensibility
 - Style of title and abstract
- Coherence with the state of the art
 - Number of references
 - Statistics of the references (e.g., reviewers might partially assess the work by checking if the work references the canonical works on the same topic)
- Respect of traditions

These factors also depend heavily on the timing at which the impact of a paper is to be predicted. If the modeling is conducted post-publication, it becomes possible to incorporate metrics such as accessibility and the Journal Impact Factor¹⁰. When the goal is to predict long-term impact, for example one year after publication, early citations can also be included as a factor. In this study, we focus on predictions made prior to publication, thereby excluding features such as early citations or journal-specific metrics.

Logos factors are particularly challenging to model, as no simple quantitative metric can adequately capture the quality of a work. While rapid progress in NLP, particularly through embeddings and transformer models, may enable substantial advances in automated assessment of scientific content, this lies beyond the scope of the present study.

3.3.3 • *MACHINE-LEARNING MODELLING*

In parallel with the mechanistic modeling presented in a previous subsection, many authors have attempted to predict the impact of papers using only measurable features and statistical or machine learning models. For example, [36] attempted to assess the influence of different features through a multiple linear regression, while [37] trained a classifier using features based solely on the past success of authors.

It should be noted that one cannot expect high predictive performance from a statistical learning model relying on only a limited set of measures of a work's ethos and logos. However, the primary objective of such modeling is not to forecast the future success of works—which remains largely utopian—but rather to enhance our understanding of the various factors influencing impact.

¹⁰The 2013 San Francisco Declaration strongly recommends against assessing the quality of a work based on its journal impact factor; however, this heuristic remains common.

4

METHODS AND DATA

DATA EXTRACTION, PREPARATION AND MANIPULATION

4.1.1 • DATA EXTRACTION

Several sources provide extensive data on academic works. The most well-known are Web of Science (Clarivate), Scopus (Elsevier), and PubMed (NIH) for medical sciences. However, these databases are not open access; for this reason, OpenAlex was preferred. OpenAlex is a free and open academic database built on the foundation of the Microsoft Academic Graph (which was discontinued in 2021). Its main drawbacks are that it can be more difficult to handle and occasionally has lower data quality.

Using the OpenAlex API, we extracted data on 26,000 works from the British Journal of Pharmacology and 306,902 works in Pharmacology.

The second extraction (works in Pharmacology) is more subjective for two reasons:

- What are the precise boundaries of Pharmacology? For example, should immunology be considered a subfield of Pharmacology?
- Once boundaries are defined, how should related works be extracted? Based on title and abstract? On journal name?

Another concern is the academic quality of the extracted works: OpenAlex includes some materials that are not peer-reviewed, such as preprints, conference proceedings, and articles published in predatory journals.

Method 1: To ensure the quality of works considered, we decided to extract all works from the top 50 journals in Pharmacology.

A subsequent bias arises from this approach: we only study works that succeeded in being published in top journals. Consequently, our dataset may not fully represent all of Pharmacology. For instance, top journals may publish a higher proportion of works with international collaborations compared to the average.

To extract Pharmacology journals, we queried the API for all journals containing a typical keyword (see list below) in their name. This yielded 1,672 journals.

Method 2: We selected all journals whose names included the following words: pharmacology, pharmacological, pharmaceuticals, pharmaceutical, pharmacy, drug, drugs.

The main traditional metrics for ranking journals are the Journal Impact Factor (JIF) and the h5-index. However, both are calculated based on works from only the last 2 or 5 years. In this study, we sought a metric reflecting the entire historical influence of journals. For this reason, we computed a “global” JIF ourselves and filtered journals to exclude those with fewer than 150 works in total.

Method 3: Journals were then ranked according to the mean number of citations received by all their works (as recorded in OpenAlex).

The final selection of 50 journals was compared with online rankings and reviewed by Dr. Péter Ferdinandy.

4.1.2 • DATA MANIPULATION

For the first dataset, its smaller size allowed us to apply certain manipulation techniques that were not feasible for the second one. Therefore, although the analysis methods are similar, the preparation and manipulation of the data differ between the two cases. For the second dataset, in particular, batching (dividing the dataset into several chunks) was required after extraction to avoid memory saturation. We used Parquet files instead of CSV files and—more critically—the Polars library instead of Pandas. Avoiding all line-by-line manipulations was essential, so only vectorized operations were applied.

Cleaning and formatting the data was also a time-consuming task. For example, when comparing contributions from companies, universities, and national institutions, companies initially appeared to be underrepresented in the top ranks of Pharmacology, suggesting that their direct contributions were modest. However, this was largely due to major pharmaceutical companies being split into dozens of subsidiaries. Standardizing affiliation names was therefore necessary. The topic modeling described below also required extensive cleaning steps.

Because of missing data for older works, some analyses begin more recently than others (e.g., from 1960 for the number of countries per work). Moreover, yearly citation data (in contrast to total citation counts from publication to 2025) is unavailable prior to 2012.

4.1.3 • FEATURE ENGINEERING

The variables extracted from the data are the followings:

EXTRACTED FEATURES	COMPUTED FEATURES
• Year of publication • Journal • Title and Abstract • Cited by count • Cited by counts year by year from 2012 to 2024 • Authors, and their institutions and countries • Distinct number of institutions and countries implied in the work • Topics, Subfields and keywords generated by OpenAlex (very low quality)	• MNCS and citation rate • Number of authors • Meta-analysis and review (True/False, based on their titles and abstracts) • Countries mean mncs and count at the time of the publication • Institutions mean mncs and count at the time of the publication • Topics, with a topic model based on the titles and abstracts (high quality) • Topic mean mncs and count at the time of publication

For all country- and institution-level metrics, a work is assigned to every country listed in the affiliations. We are therefore counting contributions rather than the works themselves. [21] uses the same method.

Addressing the age bias described in the [literature review \(4.1.3\)](#) is important. This bias is evident in our study. *Figure 1* (left) shows that, on average, Chinese contributions are much younger than European ones. As a result, they have had less time to gain recognition (or, conversely, to become obsolete), making comparisons of countries' research quality inaccurate if based solely on $N_{\text{citations}}$.

More precise metrics include the citation rate and the MNCS. The MNCS has been found to be an unbiased metric, whereas the citation rate is unbiased only if papers do not age (i.e., if attention to the paper does not decay over time). *Figure 1* (right) shows a lower correlation between the citation rate and $N_{\text{citations}}$ than between MNCS and $N_{\text{citations}}$. This indicates the presence of an age bias: MNCS provides a good compromise between $N_{\text{citations}}$, which overly penalizes recent works, and the citation rate, which overly penalizes older works. This finding is consistent with the literature.

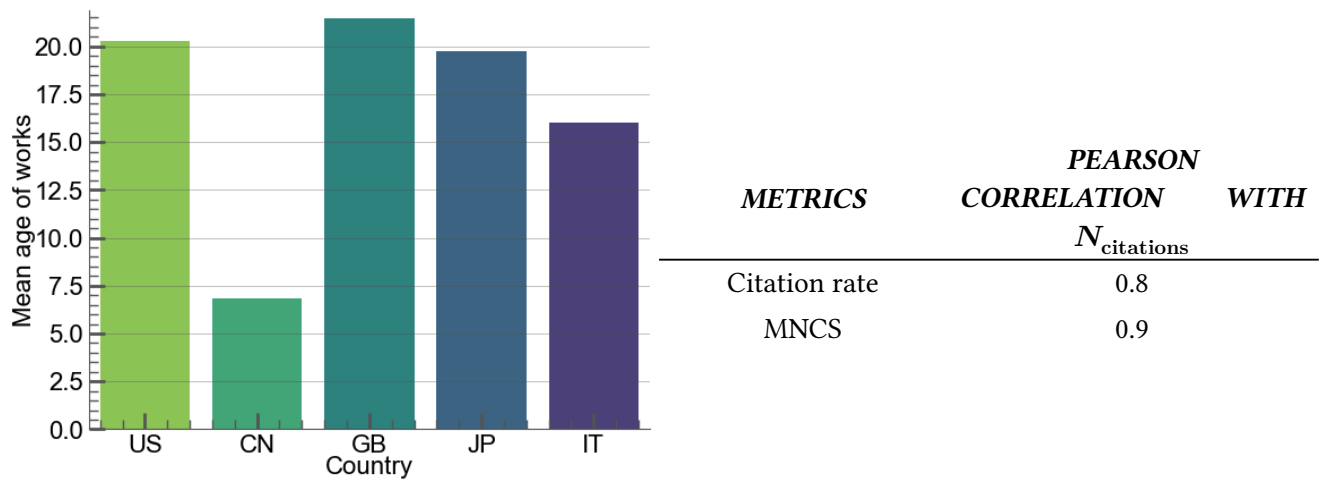


Figure 1: Mean age of the contributions according to the country (**left**). Correlation of the MNCS and the citation rate with the number of citations (**right**).

DATA ANALYSIS AND MODELLING

4.2.1 • TOPIC MODELLING

A topic modelling of our corpus of works have been made using BERTopic algorithm on the titles and abstracts. The main steps of this algorithm are:

- **Tokenization** (text segmentation into small units)
- **Semantic representation of the documents:** BERTopic uses an embedding pretrained transformer (based on the BERT language model) to convert words into semantic vectors. all-MiniLM-L6 was used for embedding. These vectors are then aggregated by document. We used Sentence-Transformers for this step called pooling
- **Dimension reduction:** the embedding space has a very high dimension. A dimension reduction is often necessary. UMAP have been chosen for this step.
- **Clustering:** once transformed into semantic vectors, a clustering algorithm is used to group documents into clusters. HDBSCAN have been chosen for this step.
- **Clusters labeling:** the most representative words are computed for each cluster. The c-TF-IDF (Class-based Term Frequency-Inverse Document Frequency) is used at this step. This algorithm simply rank words according to their frequency into the class compared with their frequency in the whole corpus. The algorithm of this step is the most basic one, it is necessary to clean the uninteresting words (also known as “stop words”) before this step.

The embedding is a major difference with previous algorithms that were only representing documents as “Bag-of-Words”, i.e. as list of words weighted according to their occurrences.

The different algorithms chosen (UMAP, HDBSCAN, all-MiniLM-L6) have been so for their performance and their widely use.

This modeling is computing and memory intensive and have been run on GPUs.

4.2.2 • COAUTHORSHIP NETWORK

The coauthorship network have only been studied for the works of the BJP due to computing limitations.

Visualizing a network remains highly subjective and sensitive to biases, as the results depend extensively on the chosen layout algorithm and its parameters. However, numerous metrics have been developed to extract objective information from a network, those are detailed in [Appendix C](#). In this work, we will use the ForceAtlas 2 algorithm for the layout but we will mostly focus on the classical networks metrics.

4.2.3 • CITATION DISTRIBUTION

Historically, the fit of a power law has often been assessed using a log-log plot of the density function of the observed distribution. Similarly, the fit of an exponential law is often assessed using a semi-log plot. In these two cases, the shape of the plot should be linear if the distribution follows the law ($y \propto \frac{1}{x^\alpha} \Rightarrow \log(y) \propto -\alpha \log(x)$ and $y \propto e^{-\lambda x} \Rightarrow \log(y) \propto -\lambda x$). The problem thus becomes a simple linear regression problem.

However, this method is not robust, especially for the tails of the distributions where data is often scarce and spanning a large range of values. Using the cumulative distribution function or a Zipf plot [11](i.e. a plot on which we replace the number of citations on the x-axis by the rank of the work in the dataset in terms of citations) instead of the density function is often more robust. However progress in computing allows more precise assessment of the fit, especially with the Kolmogorov-Smirnov (KS) test. A downside of this test is that it becomes very sensitive for large dataset (like the head of our citation distribution) and will be too exigent for such real data. For this reason we used a bootstrap strategy to compute KS test on smaller samples (of size 100) and then to aggregate the results.

Interpretation of the results of a KS test. A law fit with the data if the p-value of the KS test is above the 0.05 threshold.

In this work, we used both the log-log/semi-log plots for a first conjuncture and the Kolmogorov-Smirnov test for a more precise assessment of the fit.

The `powerlaw` library has also been used to compare different theoretical laws with (log)likelihood ratio.

Interpretation of the results of a loglikelihood ratio test. A loglikelihood ratio test allow to decide between two laws if the p-value is bellow 0.05. The best distribution is the first one if the ratio is positive and the second one otherwise.

4.2.4 • MACHINE LEARNING

The machine learning models have been trained using the `scikit-learn` library, pipelines have been used for avoiding any data leaking between the test and the train sets.

The goal is to predict the “long-term” impact of a works. More precisely, we will mainly try to predict whether a work will be part of the 50%/10% top works of its years in term of long-term impact.

The estimated period during which an article receive a particular attention from the scientific community is found to be around 10 years. We thus consider that if we rank the works published a given year according to their number of citations, the ranking will become approximately stable after 10 years.

With this framework, we will therefore consider all the old works published more than 10 years ago and try to predict whether the MNCS they have in 2025 is above or below the threshold to be part of the top 10% works.

5 RESULTS

DESCRIPTION AND EVOLUTION OF THE RESEARCH IN PHARMACOLOGY

5.1.1 • TOP AUTHORS AND INSTITUTIONS

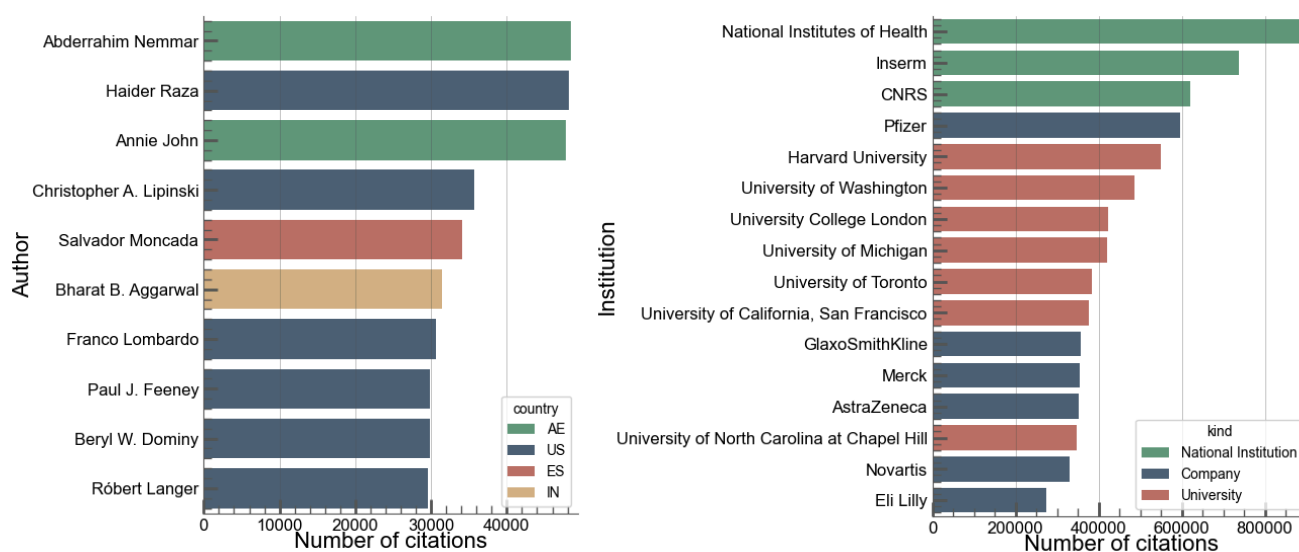


Figure 2: Top 10 authors (**left**) and top 16 institutions (**right**) in terms of number of citations in the dataset

The top 10 authors highlights a caveat of our study already mentioned: this dataset may not be representative of all the Pharmacology. Indeed, this ranking is found to be inconsistent with the existing rankings because many of the top authors are publishing in general science journals or general biology journals, and not only in journals specifically related with Pharmacology.

The institutions ranking (after merging the numerous subsidiaries of the big companies) shows a notable balance between the three kind of actors, big companies are competing with the top universities with similar global role on the field.

5.1.2 • EVOLUTION OF THE NUMBER OF PUBLICATIONS AND CITATIONS

Figure 3 shows that the number of publications per year has increased quickly from 1960 to 2020. The evolution does not follow the exponential law described in the literature. However we are considering a fix number of journals (50) and we are therefore not measuring the growth of the number of Pharmacological journals and cannot rigorously assess whether pharmacological literature is growing exponentially or not.

The number of references and citations per works (until 2010) also seems to increase fast. The number of citations per work is then decreasing, probably because recent studies lack of times to become well known. These variations of the mean number of citations with the year of publication shows the pertinence of correcting the age bias (with the MNCS). The mean size of the papers' bibliography is found to increase by one new reference every two years.

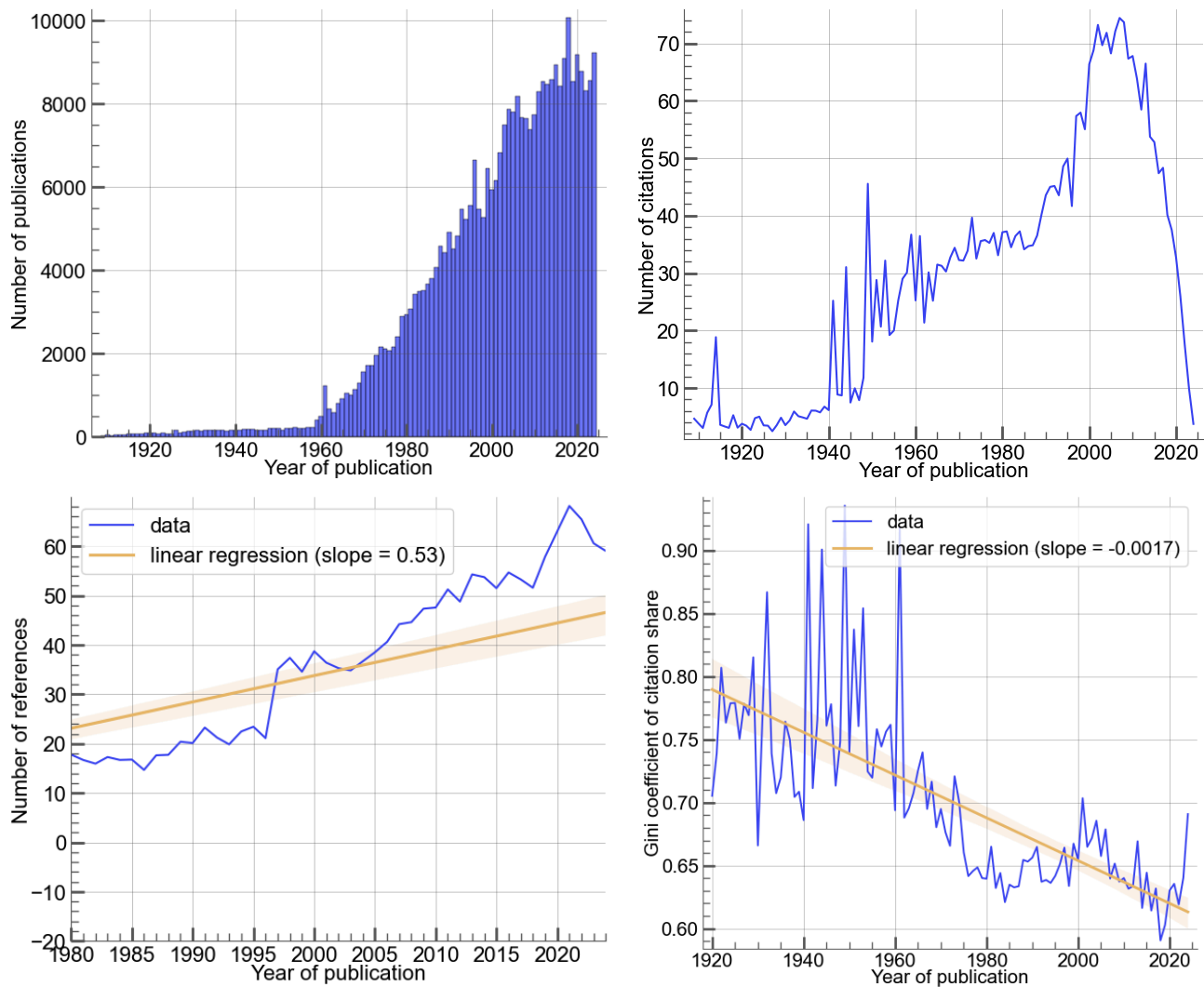


Figure 3: Evolution of the number of works published each year (**upper left**). Evolution of the number of citations per work (**upper right**). Evolution of the number of references per work (**lower left**). Evolution of the citations inequalities between papers (**lower right**).

The inequalities in terms of flow of citations across papers are found to decrease steadily (with a Pearson correlation of - 0.7). This result is opposed with the conclusion of [29] described in the [literature review](#) (citations inequalities grow when a field grow). It seems to imply an important obsolescence/renewal rate in Pharmacology: an important renewal of the important topics of research and regular breakthrough into these topics prevent a set of old papers from becoming “canonical” and oblige researchers to references mostly recent papers. We can conjecture that if we go into more granularity and observing the inequalities evolution inside a given small and mature subfield, the trend will be opposite.

$N_{\text{references}}(\text{Year of publication})$				$\text{GINI}_{\text{citations}}(\text{Year of publication})$			
Pearson		Spearman		Pearson		Spearman	
Correlation	P-value	Correlation	P-value	Correlation	P-value	Correlation	P-value
0.896	0.000	0.962	0.000	-0.714	0.000	-0.814	0.000

Table 2: Test of correlation for (number of references, year) (**left**) and (citations inequalities, year) (**right**)

5.1.3 • COLLABORATION AND THE COAUTHORSHIP NETWORK

→ “A topological transition leads the network, originally separated into many small components, to form a giant connected component.” A growth of the density coupled with a decrease in the average path length is also observed. [13]

Figure 6 shows a clear transition with a giant component that growth between 1970 and 2000 and represents around 70% of the network since the 2000s. The growth of the density is however not observed. This may be due to the small size of the dataset, that describe only a journal and not a whole field such as [13].

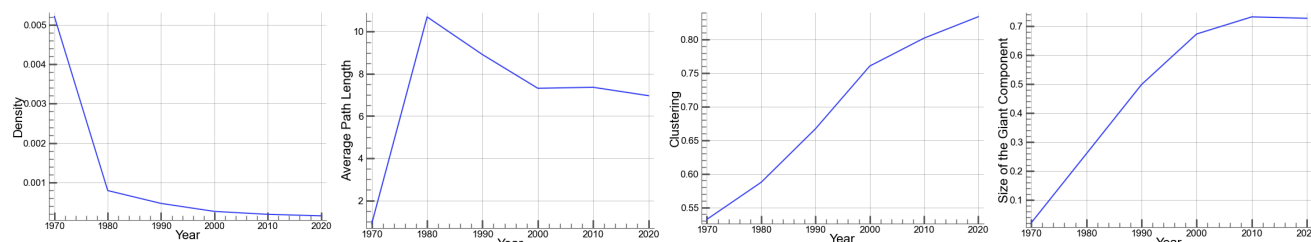


Figure 6: Evolution of the topological properties of the BJP coauthorship network: density, average path length, clustering, size of the giant component in proportion of the total size of the network

A country by country and institution by institution analysis highlights some major disparities in the evolution of the field. United States literature appears to have increased with a significantly rate than the European one, while the Chinese literature seems to growth exponentially (until the strict chinese lockdown in 2020).

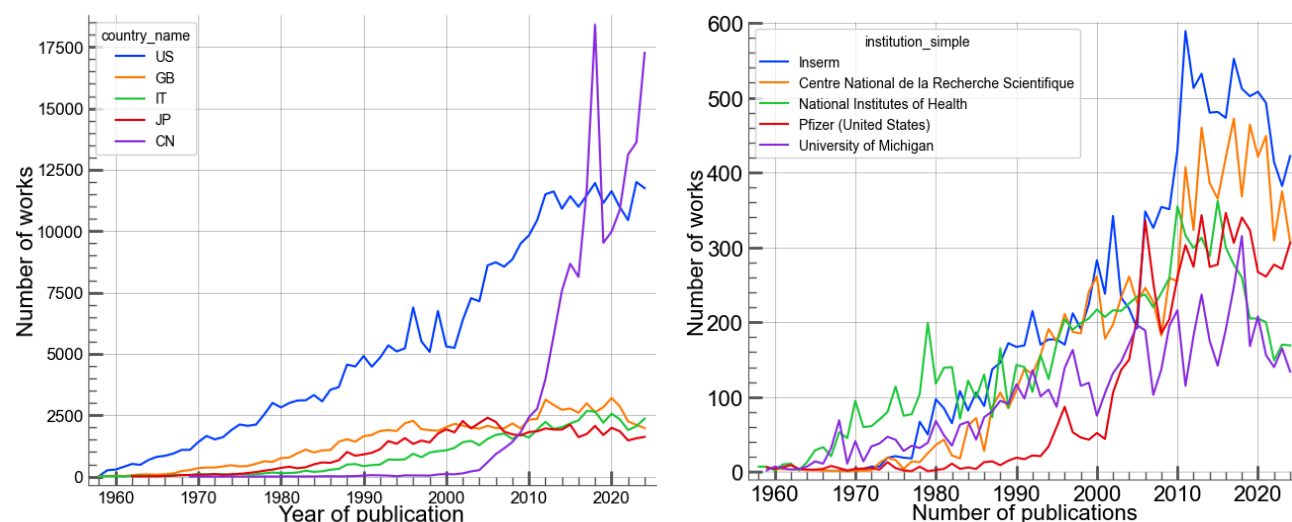


Figure 7: Evolution of top 5 countries (**left**) and institutions (**right**) in terms of number of works

5.1.4 • TOPICS IN PHARMACOLOGY

A topic modelling has been made, allowing -after cleaning- a static and dynamic study of Pharmacology’s subfields. A dynamic barchart of the top pharmacological topics over the 20th and 21st century have been made (link of the video below). Figure 8 also show the evolution of a few notable topics.

[VIDEO: EVOLUTION OF DRUGS OVER ONE CENTURY \(GITHUB\)¹¹](#)

¹¹“download/view raw”

Topics over Time

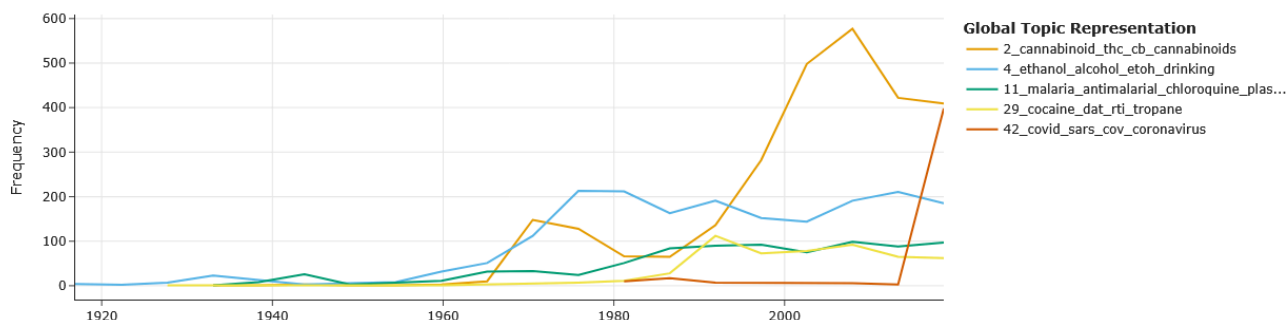


Figure 8: Evolution of 5 topics.

5.1.5 • LIFE OF WORKS

At this point of the analysis, year has always referred to the year of publication and $N_{\text{citations}}$ to the number of total citations in 2025. However it is also possible to analyse the number of citations received each year by the studies, enabling the study of the “life” of individual works, i.e the evolution of their diffusion along the years.

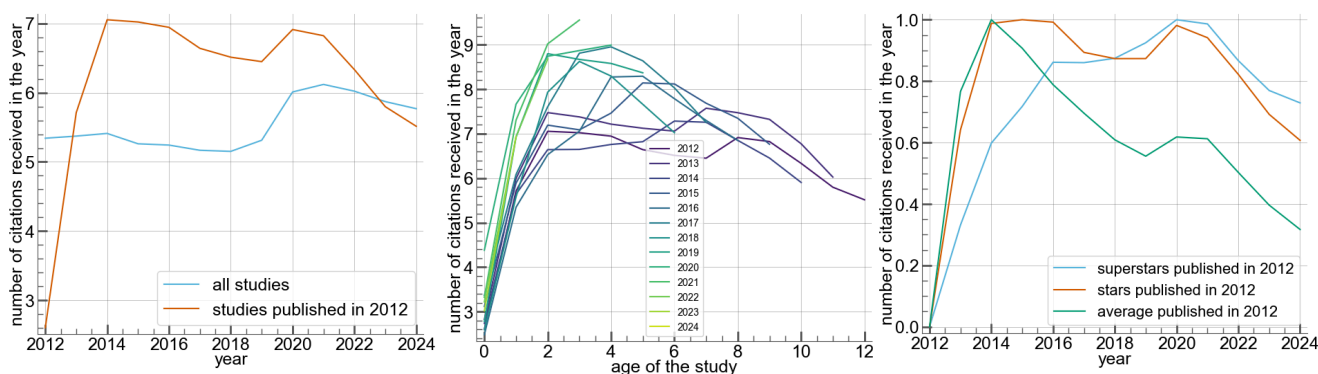


Figure 9: Mean number of citations of the works received year by year since 2012 (even the ones published before this year) and life of the 2012 works (**left**). Beginning the life of the works published since 2012 (grouped by year of publication) (**center**). Beginning of the life of the 2012 works according to their global citations in 2025 (with a min-max rescaling), “stars” are top 1% works, “superstars” are top 0.1% works (**right**).

Figure 9 (*left*) compares the mean numbers of citations received year by year by the 2012 studies compared with those of all our corpus. It appears that **new works receive more attention than the average after one year of existence and for a period of about 10 years, with an attention peak 2 years after their publications.**

Figure 9 (*center*) compute the beginning of the life of studies published between 2012 and 2024, the peak of attention 2 years after the publication is confirmed (if we exclude the COVID-19 effect).

Figure 9 (*right*) compared the life of most successful works with the life of average works (in terms of diffusion). **Top works not only receive more citations during their first years, but also retain attention during a longer period.** The attention peak of the top 0.1% works seems to be still not reached in 2019, 7 years after the publication (the COVID-19 affect exact estimation of this peak). These results reinforces the thesis of [11], [12] presented in the [review](#) (top works having a longer lifespan).

As mentioned before, a **COVID-19 effect on attention given to pharmacological works** is identified. This increase of attention is observed on all the corpus (*left*) as well as on the 2012 studies (*left, center and right*) and new works (*center*) that made particularly received particularly many early citations in 2020-2021.

DIFFUSION OF WORKS: DISTRIBUTION AND FACTORS

5.2.1 • DISTRIBUTION OF THE REFERENCES AND THE CITATIONS

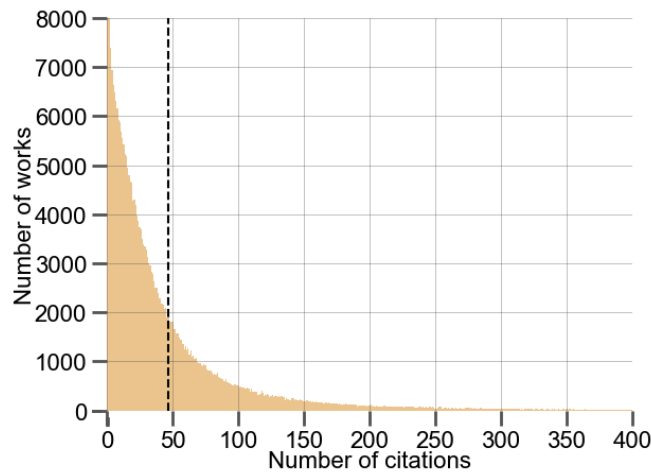


Figure 10: Distribution of the number of citations with its mean in black (48)

The citations distribution is first plotted (*Figure 10*), revealing a skewed and probably heavy-tailed distribution similarly to the literature results. The goal is then to find a law describing this distribution.

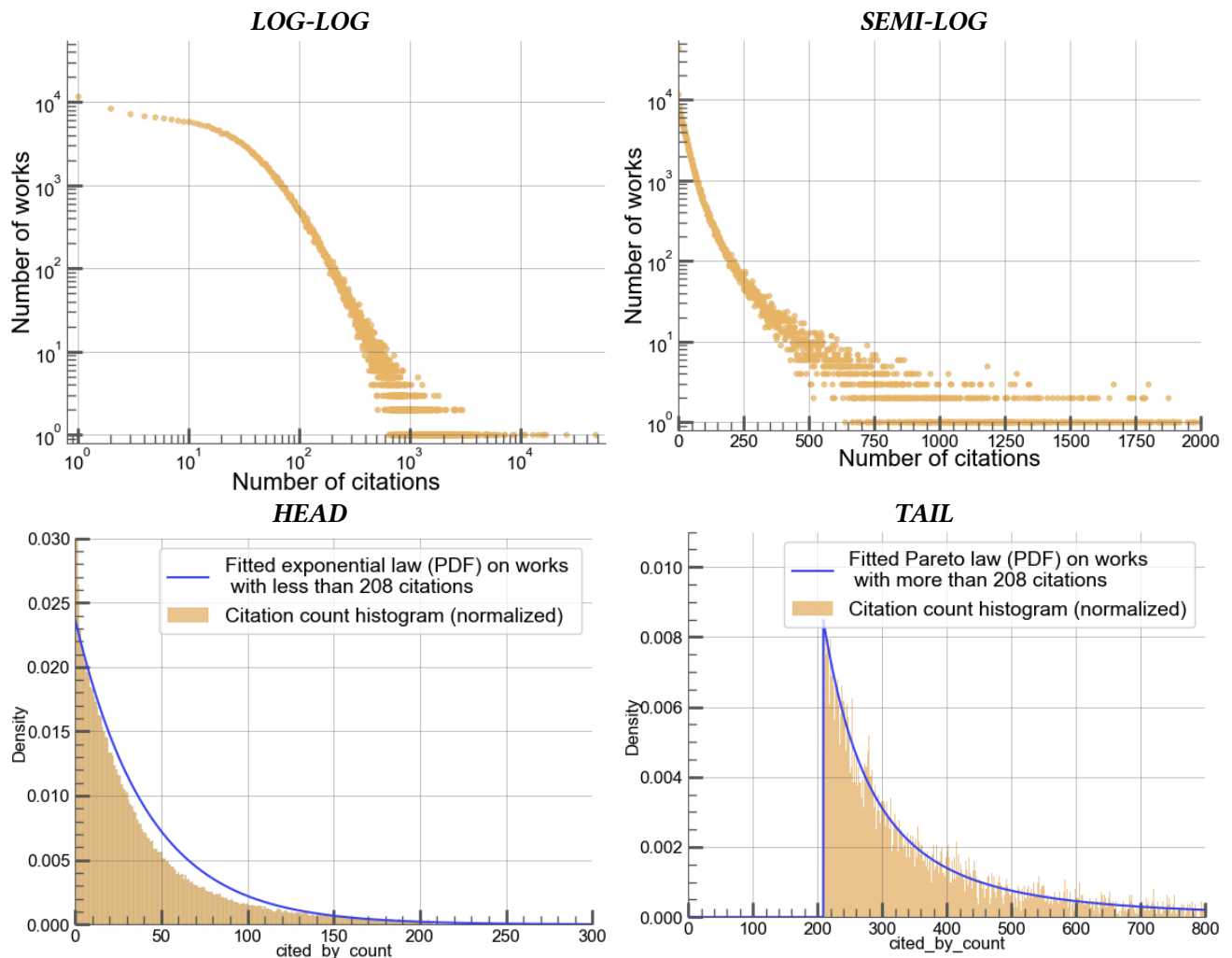


Figure 11: Distribution on a log-log scale (**upper left**) and on a semi-log scale (**upper right**). Fit of the head of the distribution with an exponential law (**lower left**) and fit of the tail with a power law (**lower right**).

The log-log distribution on *Figure 11* shows a linear asymptote for large number of citations (> 100 , even though noise become important for very large number of citations). The semi-log distribution seems in contrast to follow a linear trend for small number of citations (< 200). Even if this first description remains unprecise, it seems to indicate that the distribution is made of an exponential head and of a power tail.

Using a Kolmogorov-Smirnov test, the tail is found to follow a Pareto (/power) law for $N_{\text{citations}} > 208$. The law followed is:

$$\mathbb{P}(N_{\text{citations}} = k) \propto \frac{1}{k^{2.77}} \text{ for } k > 208$$

This result is very consistent with the [literature review](#).

Contrary to the tail, the head contain a huge number of data. This leads to further challenges (detailed in the [methods](#)). Using an approximated Kolmogorov-Smirnov test, an exponential law is found for $k < 208$ (p-value = 0.055).

To further gain in robustness, we can use a likelihood ratio to compare our choosed theoretical laws with other realistic laws.

A likelihood ratio test can be made to compare a power law and an exponential law distribution for the head. Using powerlaw package, we found a loglikelihood ratio largely in favor of the exponential law hypothesis (loglikelihood $\ll 0$ and p-value = 0.000). The exponential fit gives us:

$$\mathbb{P}(\text{citations} = k) \propto e^{-0.024k} \text{ for } k < 208$$

Interestingly, applying a likelihood ratio to compare the fit of the tail with a power law and with a lognormal law don't succeed in differencing the two (p-value = 0.8).

These results are consistent with the literature and confirm that the [dual-mechanism](#) proposed by [18] is plausible.

5.2.2 • FACTORS

All the Pearson correlations between the factors and the MNCS are presented in the [Appendix B](#).

5.2.2.1 - Does more collaboration and more references lead to more impact?

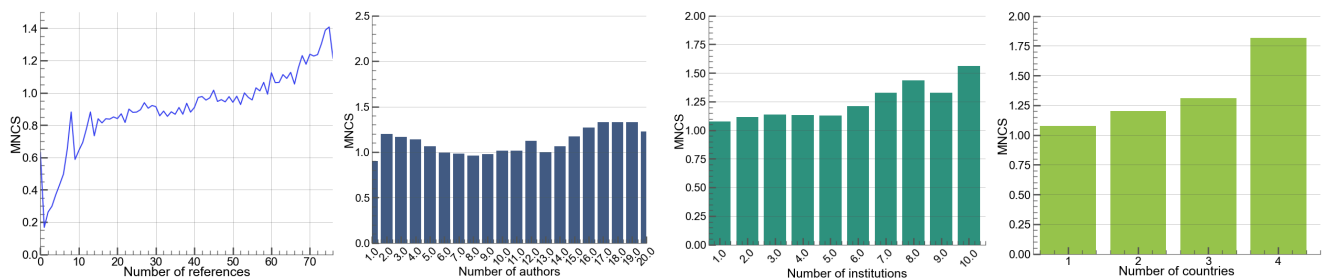


Figure 12: mean MNCS according to the number of references (**left**), authors (**center left**), of institutions (**center right**) and of countries (**right**)

No significant correlation is found between the number of authors and the MNCS (*Figure 12*). Even when considering only works with low diffusion, no correlation appears between these two variables. This seems to contradict the thesis of [19] supposing that collaboration was a main driver of [direct citations](#). A small correlation is observed for the number of institutions and a more significant one for the number of countries.

A similar graph can be made for institutions. Another linked question is whether the kind of institution (national institution, universities, big companies) influence the diffusion of a work. This “kind of institution” label does not exist in OpenAlex, therefore we only computed manually the statistics for the top institutions. The results are on Table 5 and shows significant differences with top universities having in mean 13 more citations per work than big companies.

	<i>NATIONAL INSTITUTIONS</i>	<i>UNIVERSITIES</i>	<i>BIG COMPANIES</i>
<i>Mean citations per work</i>	66	74	61
<i>Data</i>	top 3 national institutions	top 10 universities	top 20 pharmaceutical companies

Table 5: Mean diffusion of works from national institutions, universities and big companies.

5.2.2.3 - Does the topic of the work influences its impact?

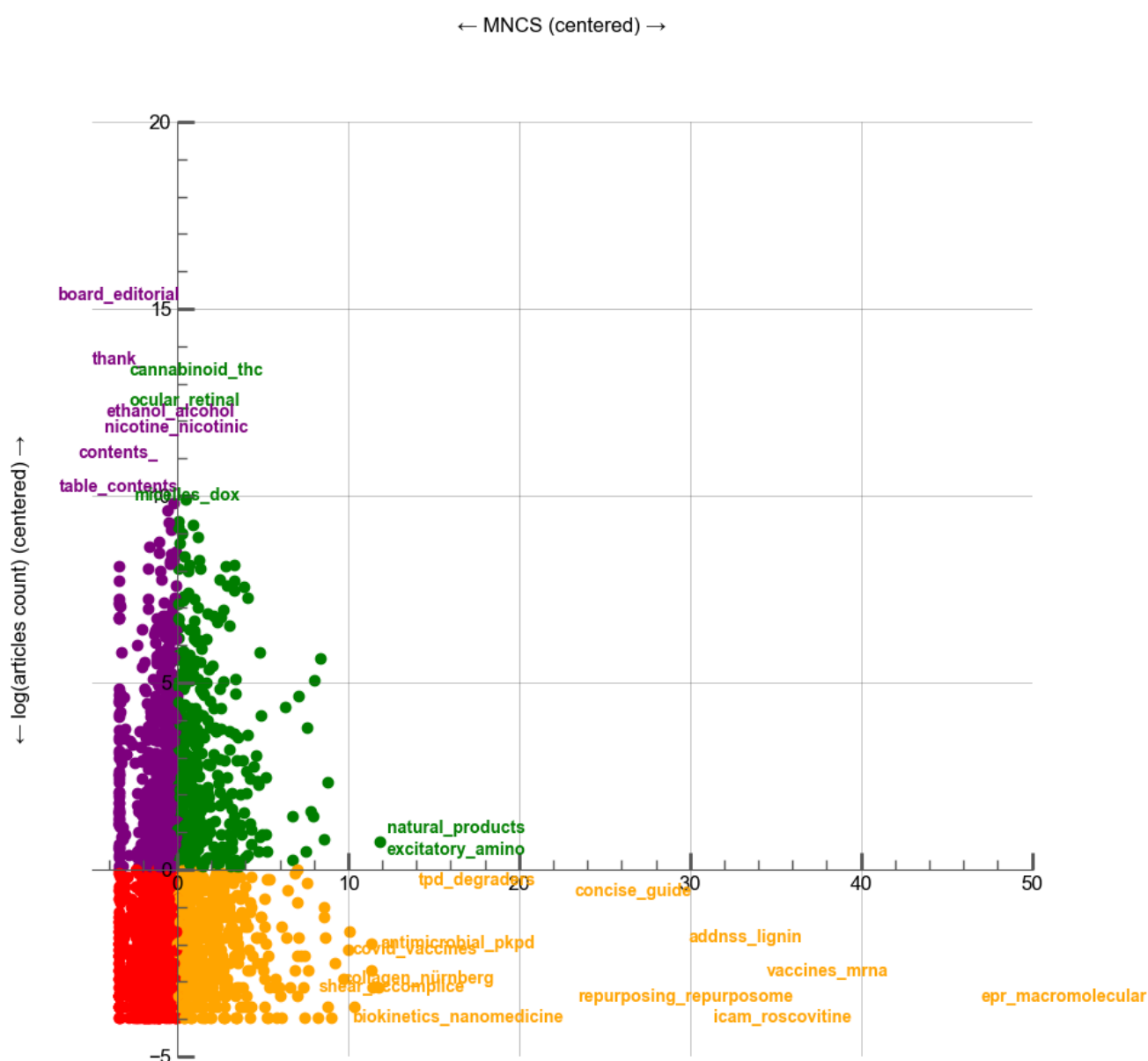


Figure 14: Topics according to the mean diffusion of their works (x-axis) and to their number of works (y-axis, log). A standard scaling have been applied.

Similarly to the country study, the topic modelling allows us to map the topics according to their size and mean diffusion, as shown on Figure 14. Such plots seems particularly interesting as they also allow dynamic study.

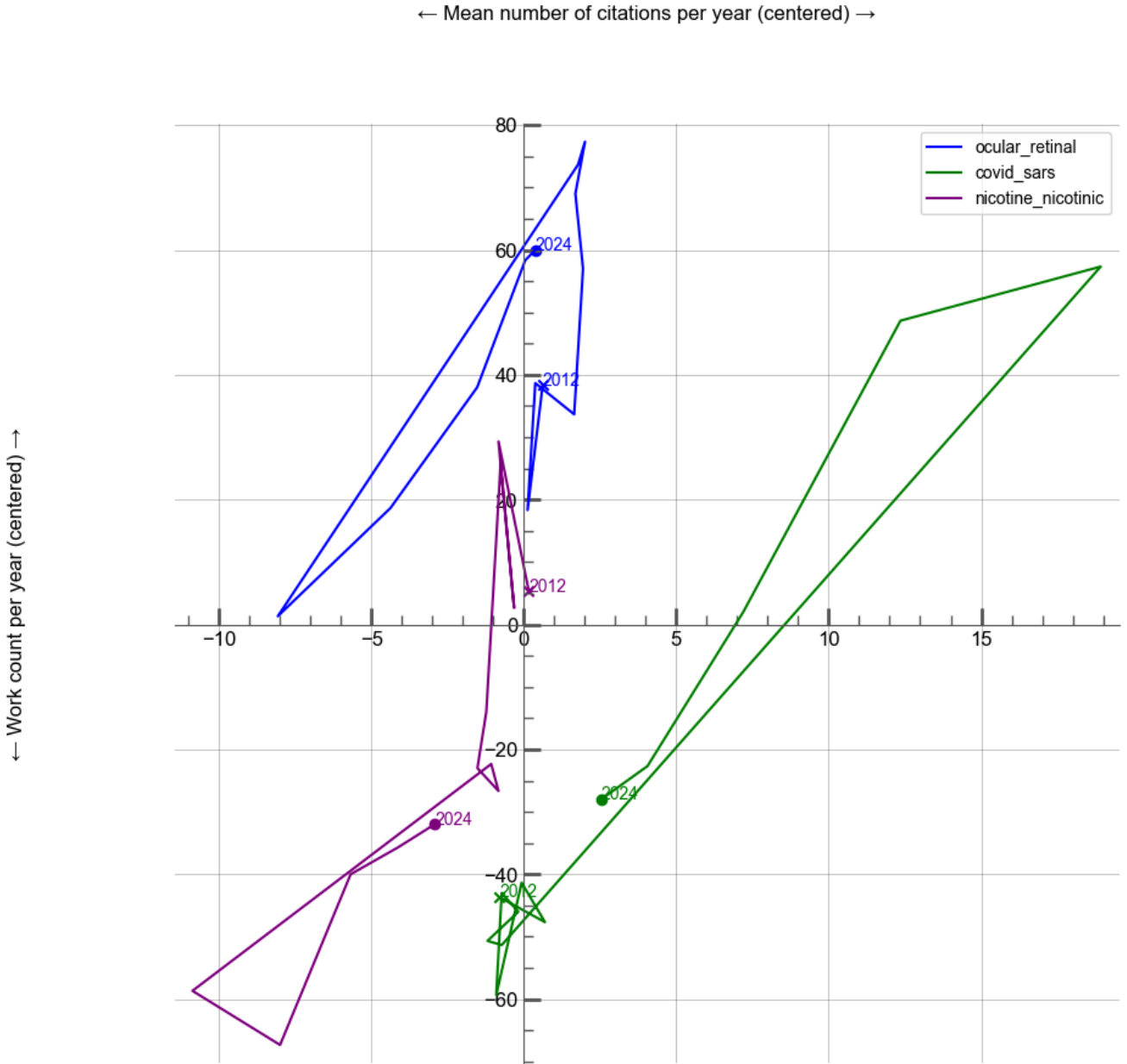


Figure 15: Topics according to the mean diffusion of their works (x-axis) and to their number of works (y-axis, log). A standard scaling have been applied.

Figure 15 shows the evolutions of 3 topics along 12 years in term of quantity of works published each years and of the number of citations received each years by the works of this topic. This graph highlights a clear correlation between the number of citations received by the works of the topic and the number of publications made in this topic (trajectories being often parallel to the $y = x$ line). Most of this correlation probably comes from the fact that more works in a field imply more citations distributed to the field.

The reciprocal causal link would be the most interesting ones: does the increase of attention given to a field cause a rise of the number of works, thus encourage scientific to study this topic of research? We would need to compare $\Delta_{y-1 \rightarrow y} N_{\text{citations}}$ with $\Delta_{y \rightarrow y+1} N_{\text{works}}$ (lead/lag analysis). With our data, we found a 0.02 Pearson correlation between the lead/lag differences. The conjuncture “scientific attention on a topic causes more production” is thus not proved. However this causal analysis remains very simplistic and we have only looked at a one year lag. Further investigations seems necessary.

5.2.3 • PREDICTING THE DIFFUSION OF WORKS

Using the features extracted from the data (listed in the [methods chapter](#), correlations available in the [appendix B](#)), we tried to predict the long-term MNCS of a paper. Pearson correlation are very low and we therefore can't expect great performances from a linear modelling (our attempts to do so lead to $R^2 < 0.1$).

A most reasonable goal is to try to predict whether a work will be part of the 50% or the 10% top works of its years in term of long-term impact.

Most significant features in terms of correlation as well as with using a feature selection algorithm (the Recursive Feature Elimination (RFE) from `sk-learn`) have been found to be:

- the mean MNCS of the authors at the time of the publication (authors prominence).
- the mean MNCS of the institutions at the time of the publication (institutions prestige)
- the mean MNCS of the topic of the work at the time of the publication (particular pattern of the topic)
- the number of references
- the review and meta-analysis status of the work.

Model	Is in top 10%				Is in top 50%			
	Accuracy	Precision	Recall	F1-score	Accuracy	Precision	Recall	F1-score
<i>Logistic regression</i>	0.89	0.70	0.54	0.55	0.67	0.68	0.67	0.67
<i>Logistic regression with the 3 principal components (PCA)</i>	0.89	0.69	0.51	0.50	0.62	0.67	0.63	0.60
<i>Random Forest¹²</i>	0.72	0.64	0.53	0.47	0.52	0.61	0.52	0.47
<i>XGBoost¹³</i>	0.90	0.78	0.51	0.50	0.67	0.71	0.66	0.64

Table 6: Results of the Machine Learning modelling

Results of our prediction attempts, using different models, can be found in *Table 6*. The high accuracies observed for top-10% prediction are due to class imbalance. Precision, recall, and F1-score were computed to address this imbalance¹⁴. The tests indicate a genuine predictive capacity (scores are higher than would be expected from random modeling), but the results remain only moderately satisfactory. Logistic regression shows the overall best performance, outperforming more sophisticated models. The low performance of the Random Forest classifier, despite Grid Search tuning of hyperparameters, is surprising. However, it has already been noted that predictions cannot be expected to perform well when features have only a weak correlation with MNCS.

¹²Best hyperparameters found with a grid search.

¹³Best hyperparameters found with a grid search.

¹⁴Macro definitions of precision, recall, and F1-score were used, i.e., the metrics are computed for each class (top 10% and top 50%) and then averaged.

6

CONCLUSION

This work provides a quantitative and historical perspective on the evolution of Pharmacology, based on more than 300,000 scientific works spanning over a century. Several key insights emerged. The field has undergone substantial structural changes, with a rapid increase in collaboration, the globalization of research, and a clear rise in the number of references per work. These trends reflect broader patterns in science, but Pharmacology also exhibits specific characteristics, such as a particularly high obsolescence rate. This suggests that knowledge production in Pharmacology is strongly influenced by both medical urgency and industrial dynamics.

The citation distribution was found to be dual, with an exponential head and a power-law tail, consistent with the cumulative advantage process described in the literature. Furthermore, the analysis indicated that collaboration per se does not strongly predict impact, whereas international collaborations and the prestige of institutions or countries correlate with higher diffusion. Topic modeling revealed heterogeneous citation dynamics across subfields.

Nevertheless, this study has several limitations. The dataset was restricted to works from the top 50 Pharmacology journals, which may overrepresent successful, internationally visible research and underrepresent more local or unconventional contributions. Additional features are lacking, such as the age and quality of references, authors' gender and social background, and the style or length of abstracts and full articles. Moreover, our focus was on predicting the long-term impact of a work before its publication; analyzing impact post-publication could incorporate features like journal statistics or early citations. Coauthorship networks were studied only for works from the BJP, and a more global analysis would be valuable.

Looking ahead, promising developments include expanding the dataset to cover a broader range of journals and publication types, improving the GGNR mechanism developed by [19], and moving from structural models to models capable of predicting the diffusion of individual papers. In this context, recent works using deep learning methods while maintaining some interpretability appear particularly interesting, such as [38].

APPENDIX A : NETWORK METRICS

Networks metrics

- **Density:** $\frac{\text{Number of edges}}{\text{Max number of possible edges}}$
- **Average shortest path length**, often to measure the **efficiency** of the network, i.e. its ability to diffuse information quickly).
- **Diameter:** longest of the shortest path length between nodes
- **Assortativity:** tendency of nodes to be connected with “similar” nodes, e.g. to be connected with nodes having the same degree (correlation between connexion and degree). An assortative network is a network where high degree nodes tends to connect with other high degree nodes. The attribute to assess the similarity is here the degree but can be different (country...)

Nodes metrics

- **Clustering coefficient:** It assesses local density. $\frac{\text{Number of connections between the neighbours}}{\text{Max number of connections between the neighbours}}$
- **Centrality:** Aims to measure the importance of a node in the graph. Dozens of centrality metrics exists and their use depends of the context of the study. The most popular are:
 - **Degree Centrality:** simply degree of the node
 - **Closeness centrality:** average length of the shortest paths between the node and all the other ones.
 - **Betweenness Centrality:** quantify the number of time a node acts as a bridge, a proxy along the shortest path between two nodes

Nodes metrics derivated from the citation network

- **Pagerank [39]**

It is defined as:

$$PR(A) = (1 - d) + d \left(\sum_{t \in \{\text{pages citing A}\}} \frac{PR(t)}{C(t)} \right)$$

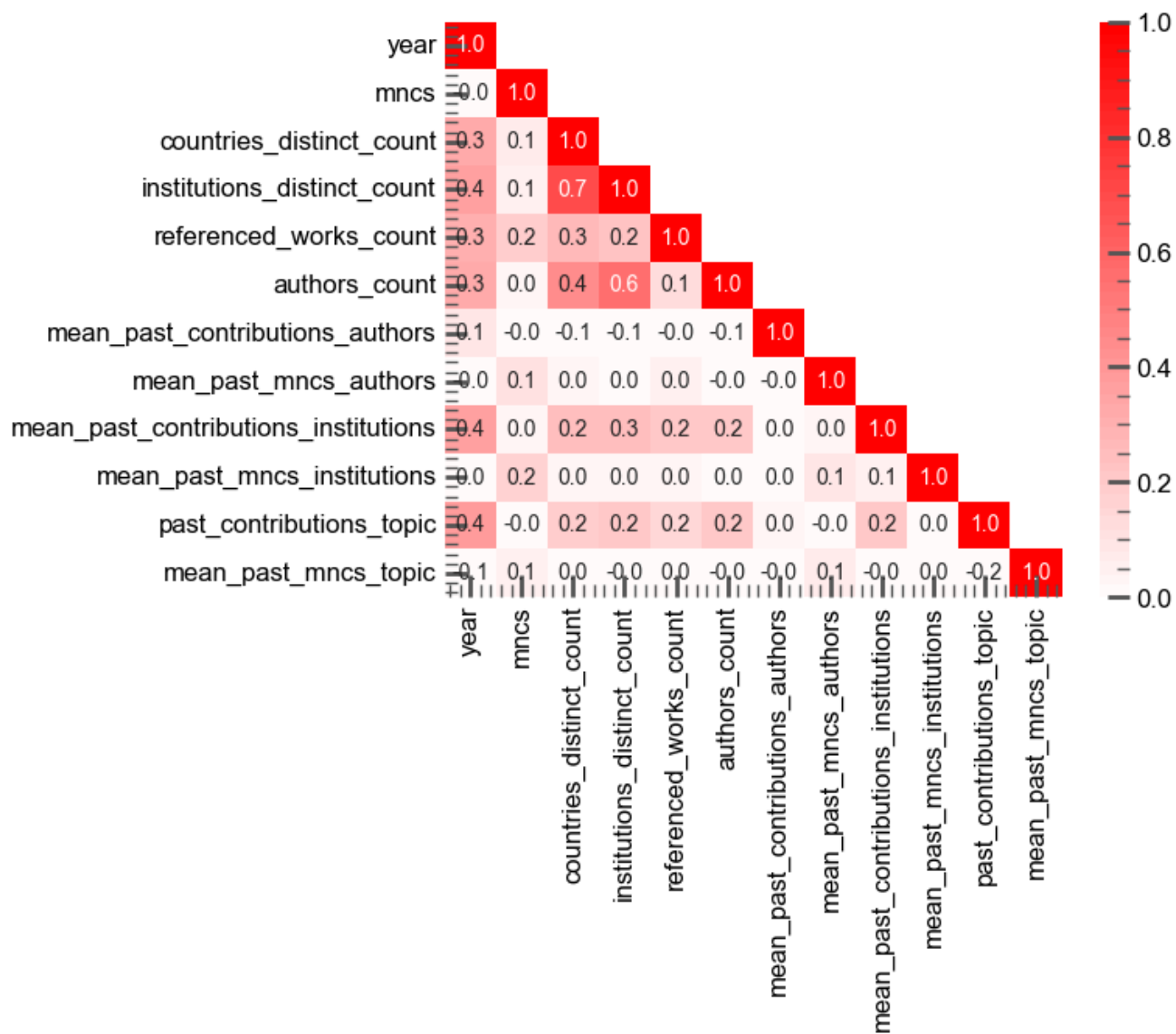
With :

- PR the pagerank of the given page,
- d a damping factor,
- C the number of citation made on the given page.

When $d = 1$ (and if the graph is strongly connected and aperiodic), the pagerank values correspond to the principal eigenvectors of the normalized adjacency matrix.

Explanations of Brin and Page: [39] “We assume there is a “random surfer” who is given a Web page at random and keeps clicking on links. never hitting “back” but eventually gets bored and starts on another random page. The probability that the random surfer visits a page is its PageRank. “

APPENDIX B : HEATMAP OF CORRELATIONS



GLOSSARY

Term	Definition
SciSci	Science of Science
MNCS	Mean Normalized Citation Score, defined as $\frac{N_{\text{citations}}}{N_{\text{citations, year, topic}}}$ where $N_{\text{citations, year, topic}}$ is the mean number of citations received by the works published the same year in the same field. In our study, it is mainly define as $\frac{N_{\text{citations}}}{N_{\text{citations, year}}}$ as we already work on one specific field. This notion is detailed in the literature review .
Citation rate	Mean number of citations received per year $\frac{N_{\text{citations}}}{\text{age}}$. More details in the literature review .
Network metrics	Cf. Appendix A.
GGNR	Cf. the Methods .
Accuracy	Percentage of correct predictions
Precision	Proportion of true positive results in all positive predictions
Recall	Proportion of true positive results in all actual positive cases
F1-score	Harmonic mean of precision and recall
Zipf Plot	Cf. the Methods .
Pearson and Spearman correlations	Describe the linear (Pearson) and monotone (Spearman) relationship between two variables
KS test	Cf. the Methods .
Matthew Effect	Cf. the review .

PLANNING OF THE INTERNSHIP

Date	Task
02/06 - 06/06	Literature review
09/06 - 13/06	Meeting my cotutors, literature review
16/06 - 20/06	Data analysis
23/06 - 27/06	Getting familiar with OpenAlex and data manipulation, data extraction of the British Journal of Pharmacology (BJP) works
30/06 - 04/07	Network analysis, getting familiar with Gephi, Graphistry, Networkx
07/07 - 11/07	Country, institution and authors analysis, attendance to a workshop on Science of Science
14/07 - 18/07	Topic modelling and analysis
21/07 - 25/07	Further data analysis, Presentation of my work to Péter Ferdinandy (and preparation of the presentation), director of the BJP and vice-rector of the medical university of Budapest (Semmelweis Egyetem).
28/07 - 01/08	Extension of the work to a new dataset Pharmacology, summer school organized by the lab
04/08 - 08/08	Data analysis, machine learning and topic modelling on the new dataset, summer school organized by the lab
11/08 - 15/08	Improvement of the literature review, report writing
18/08 - 22/08	Report writing, improvement of some aspects of the study

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